

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.20)

Metric

$$ds^2 = \left(-\frac{r - 2m(u)}{r} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 d\theta \otimes d\theta + r^2 \sin^2(\theta) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (u, r, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{m(u)}{r} + \frac{1}{2} \right) du + (1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) du + \left(-\frac{m(u)}{r} + \frac{1}{2} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(\frac{i\sqrt{2}}{2r\sin(\theta)} \right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(-\frac{i\sqrt{2}}{2r\sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{m(u)}{r} + \frac{1}{2}\partial_r$$

$$\delta = \frac{\sqrt{2}}{2r}\partial_\theta + \frac{i\sqrt{2}}{2r\sin(\theta)}\partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2r}\partial_\theta - \frac{i\sqrt{2}}{2r\sin(\theta)}\partial_\phi$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2r} - \frac{m(u)}{r^2}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2} \cos (\theta)}{4 r \sin (\theta)}$$

$$\beta = \frac{\sqrt{2} \cos (\theta)}{4 r \sin (\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{m(u)}{2 r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{\frac{\partial}{\partial u}m(u)}{r^2}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{m(u)}{r^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "